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Mapping the Potential: Generative AI and Public Sector Work

**Using time use data to identify opportunities for AI
adoption in Great Britain's public sector**



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Introduction

An emergent and rapidly evolving technology, artificial intelligence (AI) in its varying forms is reconfiguring the operational landscape of government (van Noordt and Misuraca 2022; Vogl et al. 2020). From enabling new forms of civic engagement to supporting more robust fraud and error detection (Giest and Klievink 2024; Government Digital Service 2025), AI and other automated technologies stand to transform the public sector. In late 2023, preliminary analysis of AI's potential impact on public sector productivity levels predicted gains “worth billions” (National Audit Office 2024). However, despite the promising opportunities presented by this technology, there remains much work to be done in terms of actually realising the benefits. A recent review of the current state of digital government in the UK showed that the public sector's digital foundations are strained and have not developed at pace, resulting in over £45 billion of public spend lost in unrealised productivity benefits per year (Department for Science, Innovation and Technology (DSIT) 2025b).

In January 2025, the UK government released the AI Opportunities Action Plan (Department for Science, Innovation and Technology (DSIT) 2025a), a roadmap which acknowledges the challenges posed by the current digital foundations and outlines the government's commitment to overcoming these by harnessing AI for driving economic growth, improving productivity, and enhancing outcomes for citizens. The roadmap puts forward several actions for tackling the barriers facing the effective deployment of AI across the UK's public sector, including a ‘Scan > Pilot > Scale’ strategy to AI adoption – an approach which emphasises the need for developing a robust “understanding of AI capabilities mapped to their highest impact challenges and opportunities”.

In this report, we aim to contribute to this approach, by outlining the findings of a measurement exercise undertaken to assess the extent to which public sector activities are suited for generative AI² use. The report is part of a wider collaboration between The Alan Turing Institute, the Department for Transport (DfT) and the Office for National Statistics (ONS), and represents a novel contribution to the mapping of AI's capabilities in the context of the public sector. Here, we draw on the data from the ONS Public Sector Time Use Survey (PSTUS) (Office for National Statistics (ONS) 2024) to map public sector work activities to current state-of-the-art generative AI capabilities, with the aim of identifying potential opportunity areas for adoption. We also make use

² The term generative AI refers to a subset of advanced AI technologies that can generate new textual, visual, and auditory outputs based on 'prompts' or user inputs. Generative AI tools have seen tremendous uptake - since its release in November 2022, for example, ChatGPT has amassed over 200 million weekly users (Reuters 2024), and its capabilities have been integrated into various existing tools, software, and workflows.

of the PSTUS estimates of average time spent executing these activities to provide a first estimate of how much worked time in Great Britain's public sector is spent on activities that could make use of generative AI.

According to our work, approximately 41% of public sector time is spent on activities that could be supported through the use of generative AI. This figure is varied across different areas of the public sector, ranging from a high of 49% in Education to a low of 33% in Healthcare. The figure is lower for frontline workers (38%) than it is for non-frontline workers (47%). These findings demonstrate the potential supporting role that generative AI could play in freeing up valuable public sector time. For example, the PSTUS reveals that the average public sector worker spends 30 minutes per day on emails – here, a generative AI tool could support with the initial drafting of emails, for example. While these drafts would still need to be reviewed and edited before being sent off, a generative AI tool can still support a time-pressed public sector worker by streamlining one or several aspects of the process.

In what follows, we provide an overview of our methodology followed by an acknowledgement of the exercise's limitations. We then present the findings of our measurement exercise and conclude with a discussion of these findings and potential pathways forward for future research.

Methods

In this report, we make use of data from the recently released PTUS conducted by the Office of National Statistics. In this survey, public sector workers living in Great Britain completed a diary where they recorded how they spent their time during a typical working day (including work and non-related activities). Participants recorded how they spent their time using 91 activities. These activities were grouped into 20 categories; 18 of the work specific activities were divided according to whether they were considered sector-specific (those which would typically be undertaken within one area of the public sector; for example, patient management in the health service) or non-sector specific (those which are common amongst different areas of the public sector; for example, writing and sending emails and attending, organising, or chairing meetings and events. This also includes specialised activities such as data analysis or writing up medical notes). The remaining two activity groups were non-work related and categorised as ‘other’. A full list of these activities, as well as their categories, can be found in the [Appendix](#).

In order to assess the potential for applying generative AI to both sector and non-sector specific activities, we categorise the work activities from the PSTUS according to the extent to which generative AI could be used to complete them. This classification was based on a rubric developed by Eloundou et al., (Eloundou et al. 2024) which was grounded in established ‘task-based’ methodologies from the field of labour economics. The task-based approach, also commonly referred to as the task-model, was initially developed by labour economists to support the study of the impact of technological change on occupations (Autor, Levy, and Murnane 2003). In this approach, occupations are regarded as bundles of tasks and are organised around the skills required to complete these tasks. Tasks are classified as broadly either ‘routine’ – made up of repetitive, sequential actions – or ‘non-routine’ – made up of complex, non-standardised actions. ‘Routine-task biased technological change’ (RBTC) was a theory stating that routine tasks appeared to be more susceptible to computerised automation than non-routine tasks (Walo 2023). The approach provides both a theoretical and practical framework for assessing the differential impacts of technology on human tasks.

As technology has evolved, iterations of the task-model emerged which aimed to account for the emerging capabilities of AI (Frey and Osborne 2017; Brynjolfsson, Mitchell, and Rock 2018; Hertog et al. 2023). Serving as our conceptual groundwork, the task-model approach informed our analysis of the potential impact of generative AI on public sector work activities. This builds on previous work carried out which looked to

assess the potential for AI and automation technologies on government services (Straub et al. 2024). From here, we developed our analytical rubric. The rubric is a modified version of two “automation exposure” rubrics developed by Eloundou et al., to measure the extent to which tasks are “exposed” to the development of generative AI (Eloundou et al. 2024).

Our rubric also considers a separate taxonomy of tasks developed to assess the impact of new technologies on work by also considering the methods and tools of work, which refer to (1) the physical, material methods and tools that may be undertaken as part of the completion of an activity, and (2) the way an activity may be organised around social interaction (Fernández-Macías and Bisello 2022). Finally, the rubric also considers a task’s error tolerance, and the potential consequences impacted communities may face if action is taken based on an erroneous outcome. Together, these dimensions attempt to capture the potential that generative AI has to support the completion of a given task (in keeping with the terminology of the PSTUS, we henceforth use the term ‘work activities’ in reference to tasks). Following Eloundou et al., we assess the different levels of exposure to generative AI that a given work activity may have based on a) its amenability to the technology, b) the impact this access may have on the amount of time spent on the execution of the activity, and c) the quality of its output (Eloundou et al. 2024). The degree of exposure serves as a metric for determining the potential for generative AI to be applied to a work activity.

The rubric begins by providing a hypothetical assessment statement wherein coders are asked to consider a public sector work activity and to assess its potential level of exposure to generative AI based on a number of guiding prompts. Five categories of exposure are provided on a scale from E0-E4, each then corresponding to a score which indicates the potential for generative AI adoption. On the one end, the category E0 indicates ‘no exposure’, meaning no aspect of the work activity is amenable to the use of generative AI, and no potential time reduction can be attained. On the other end, the category E4 indicates ‘full exposure’, meaning the activity is fully amenable to generative AI adoption, all aspects of the activity can be executed by the technology to a quality equivalent to that of a human without requiring supervision, and hence up to 100% of the time currently spent on the activity can be reduced. Following Eloundou et al., these exposure categories are then numerically expressed as a ‘score’ on a scale from 0-1 to facilitate the process of analysis, giving an approximation of generative AI’s potential impact on a work activity (Eloundou et al. 2024). The full rubric, alongside the assessment statement and further details on the values of the numerical schema, can be found in the [Appendix](#).

For the purpose of this project, we only considered work activities from the PSTUS that took place during working hours. We therefore excluded two main activity categories relating to activities such as socialising or sleeping that took place outside of work. There were also a few catch-all “other” ungrouped activities (such as “Other healthcare”) which we did not code directly. Hence, in total, we included 80 of the 91 activities in our analysis, spanning across 18 main activity categories. Two members of the research project team independently coded each of these 80 activities in terms of their potential exposure to generative AI, following the instructions set out in the rubric. In total the coders agreed 82% of the time on their codes. A further 16% of codes were separated by just one point (for example, the same activity was given E1 by one coder, and E2 by the other). For the codes where there was a disagreement, a final code was resolved through discussion with the two coders and the project lead.

These codes were then translated into an approximate generative AI exposure score, following the schema set out in Eloundou et al’s., paper, giving each of the 80 activities a score between 0 and 1 (Eloundou et al. 2024). These activities were grouped according to their main activity category, and an average score of all activities within the specific category was assigned. For example, the “Communication” category includes 7 activities, ranging from “media and press duties” (scored 0.5) to “writing letters” (scored 0.75). The final score for this category was therefore the average of the scores for all 7 activities.

Limitations

It is worth noting that, while aiming for the rubric to be as accurate as possible, there are limitations with our approach. Firstly, the coding exercise was undertaken by researchers from within the project team who, while they are specialists in the area of government digitisation, do not directly work in these areas within the public sector. While the rubric has been developed with a public sector context in mind, the process of assessing work activities and assigning exposure codes to them is ultimately a subjective process (as evidenced, for example, by instances of intercoder disagreement) that is impacted by the positionalities of the research project team (Bourke 2014). The coding exercise would therefore undoubtedly benefit greatly from direct engagement with public sector workers, or individuals coming from a public sector background.

Secondly, a key challenge faced during the coding process was the ambiguity of certain work activities. Work activities like ‘data entry’ and ‘taking notes’ are relatively clearly defined, and as such could be coded with ease using the rubric, with no disagreement between the coders. However, other work activities, such as ‘safeguarding’ and ‘military duties’ were relatively broad, and might encompass multiple sub-activities, each with opposing degrees of suitability for generative AI application. Further, although some activities clearly require an element of human interaction, the degree and specifics of this interaction were not always immediately clear to the research team. For example, the activity ‘teaching and assisting learners in the classroom’ is relatively high-level and does not necessarily capture the breadth or quality of interaction undertaken by public sector workers completing this task. As such, ambiguous work activities were difficult to interpret precisely in terms of their exposure to generative AI. In the absence of clarity surrounding the exact content of a work activity, assigned codes should therefore be treated as approximations. In future iterations of the PSTUS, subdividing broad work activities further and providing a brief description of what each activity entails would be immensely valuable, and would aid the robustness of mapping efforts such as this one greatly.

Finally, as outlined above, the codes assigned to activities are then assigned a score from 0-1 in order to aid in the analysis of generative AI’s potential impact, a process which also entails another level of simplification. For the purposes of analysis and reporting, each score is represented by its upper bound – in other words, an exposure score of 0.75 is interpreted as 75% of a work activity being directly exposed to generative AI, as opposed to anywhere between 50-75%. This simplification means that

these findings might in some instances over-estimate the potential for exposure to generative AI. Finally, our measure does not take into account

Results

Findings from the PSTUS show that the average public sector worker works 8.3 hours per day³. Based on our model, approximately 3.4 hours (out of the 8.3 hours) are potentially exposed to generative AI (41%). The distribution of exposure scores was relatively even between the categories E0 – E3, with approximately 20-25% of the 80 activities falling in each bracket. Only four activities (approximately 5% of the total) were given the E4 score of full exposure, which indicates that the coders believed a state-of-the-art generative AI model could fully execute the work activity to a quality equivalent to that of a human, and that we could see a reduction of up to 100% of the time spent on the activity. These were 1) Organising meetings, meeting administration; 2) Creating or updating an appointment or booking; 3) Completing or processing forms; 4) Creating or updating records, databases, case files, or similar documents.

Sector	Total hours worked	Generative AI exposure (hours)	Generative AI exposure (%)
All	8.3	3.4	41%
Healthcare	8.7	2.9	33%
Police	8.8	3.6	41%
Local Government	8	3.4	43%
Prison service	8.4	3.6	43%
Courts and tribunals service	7.9	3.4	44%
Probation service	8.2	3.7	45%
Other civil service organisations	7.9	3.8	47%
Secondary education	8.4	4.1	49%
Primary education	8.2	4	49%

Table 1: Total hours worked and potential generative AI exposure, expressed as hours and percentages. Results are broken down by service sector.

The 80 work activities analysed are undertaken with varying frequency and to varying degrees by public sector workers based on their areas of operation. Hence, exposure scores vary by service sector (see Table 1), ranging from a low of 33% of all worked time in healthcare to a high of 49% in both primary and secondary education, as shown in Table 1. There are a number of different variables that drive the difference in exposure scores across service sectors. These differences reflect variation in the nature of the different activities undertaken by those within the sectors, the degree of reliance on human interaction, the activities' error tolerance, and the need for physical

³ Note, this average excludes short working days below 3 hours.

manipulation of the surrounding environment – all dimensions which are captured within the rubric, and which determine the final exposure score. In healthcare, activities like “treating patients”, “observing patients” and “supporting patient care” require a high degree of human interaction, have low room for error, and high individual tailoring, making them less exposed to generative AI application. However, we might expect to see different results within healthcare if we were to focus on a different AI application, such as diagnostic or perceptive AI systems, which can – for example – support with activities like evaluating scans more effectively than generative AI applications can. In education, on the other hand, there is a sizeable share of more structured work activities such as “lesson preparation” and “reporting”, all of which have a high generative AI exposure score. The same pattern can be observed across all sectors: tasks that have a low error tolerance, require complex judgement, and higher levels of human interaction are less exposed to generative AI.

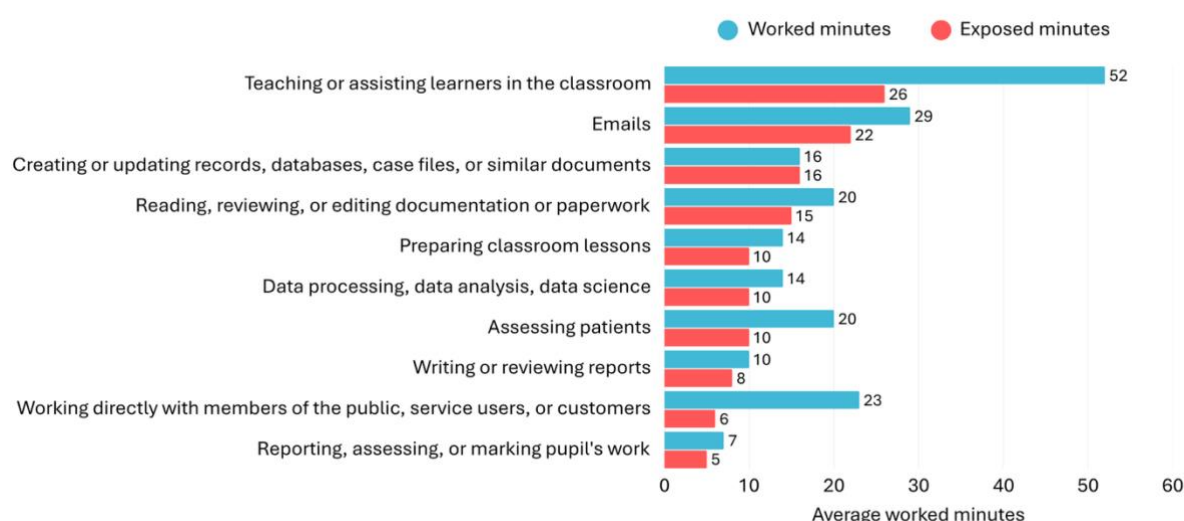


Figure 1: Top 10 activities with the highest level of exposure, expressed in minutes (overall for the average public sector worker)

While exposure scores were similar between managerial (43%) and non-managerial (45%) roles, they differed between frontline and non-frontline workers; 38% of frontline workers’ activities were found to be potentially exposed to generative AI, compared with 46% of non-frontline workers’ activities. According to the PSTUS, frontline workers participate in activities necessitating interpersonal interaction at a higher rate than non-frontline workers. These could include activities such as ‘staff management’ and ‘staff related administration’ as well as ‘other public service duties’ which include ‘social care activities’, ‘working directly with members of the public, service users, or customers’, and ‘housing or welfare related activities’. The rubric – and indeed wider literature – identifies work activities which require interpersonal interaction as being less amenable to generative AI adoption in general (Acemoglu and Autor 2011; Brynjolfsson, Mitchell, and Rock 2018; Webb 2019). While the nature and degree of

interpersonal interaction undoubtedly varies based on sectors and actors, frontline workers within the public sector are often uniquely placed to directly enact policy, regulation, or decisions with a degree of necessary discretion and human judgement. Decisions made within the public sector often directly impact individual citizens' liberties, wellbeing, and access to basic services. As such, the elements of human judgement and discretion that are part and parcel of interpersonal interaction in this sector are vital, resulting in lower exposure scores given to activities with these elements.

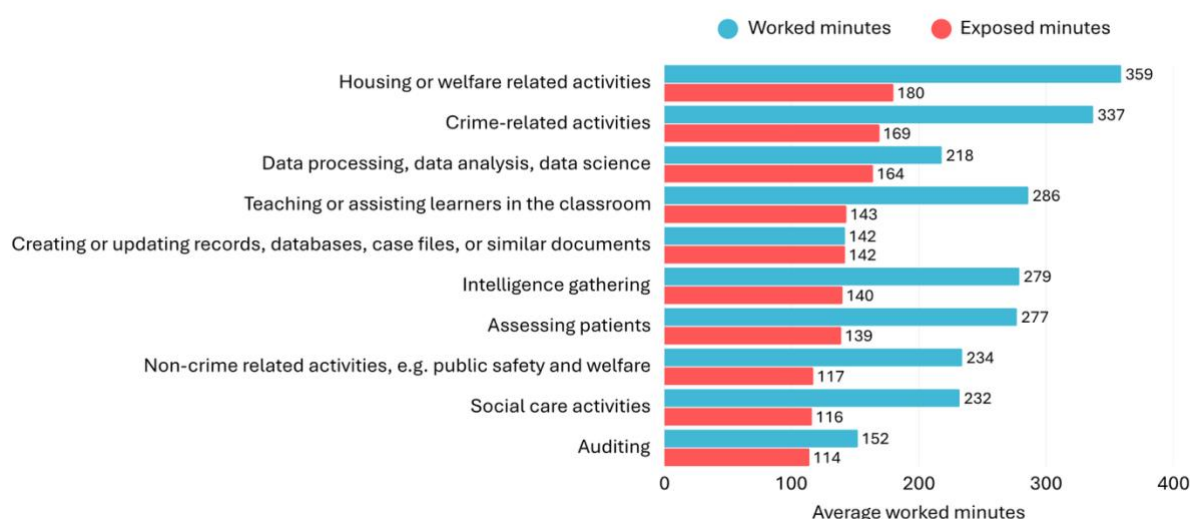


Figure 2: Top 10 activities with the highest level of exposure, expressed in minutes (by those civil servants who participated in them)

Of the top 10 work activities with the highest levels of exposure (see Figure 1), our results show that, on average, 50% or more of the time spent on these activities has potential for generative AI adoption. Some of these apply across the public sector. Public sector workers spend an average of 30 minutes per day on emails – a timeframe which, according to the methodology outlined above, could potentially be reduced by more than 70% with the use of state-of-the-art generative AI models. Similarly, the average 15 minutes per day spent on creating or updating records, databases, case files or similar documents could be almost entirely reduced. While these numbers are small in absolute terms, when aggregated across the public sector they could result in significant savings⁴. There are even more significant savings when we look at the average activities for the public sector workers who participated in them (Figure 2). For

⁴ Note, in the first wave of data collection, around half of the PSTUS sample has been taken from the ONS OPN survey, and can be considered as a randomised sample across all public sector workers. Additional sample has been collected through calls for volunteers from public sector organisations, and hence areas such as crime and justice have been oversampled due to this sampling strategy. The first wave of the pilot was made up of 1,513 respondents from across the public sector in Great Britain, with a total of 3,005 diaries. These were then weighted to the public sector population using the Annual Population Survey. For a more detailed breakdown of the sampling for this study, please see [Section 9](#) of the time use survey publication.

example, public sector workers who carry out social care activities spend an average of approximately 230 minutes on them, of which nearly half could be supported by generative AI (highlighting the significant administrative burden social care workers face). Meanwhile, a person working in the education sector spends an average of approximately 100 minutes per day preparing classroom lesson plans – this could potentially be reduced by up to 75% with the use of generative AI.

Discussion

Prior research has found that generative AI use appears to already be widespread in the UK's public sector, though a systematic and robust approach to adoption remains lacking (Bright et al. 2025). The aim of this project has therefore been to contribute to the growing evidence-base which can be used to inform the adoption of this technology in the public sector. By mapping the capabilities of generative AI technologies to work activities undertaken across the public sector, we measure the proportion of these activities that are exposed to the technology and provide estimates for the potential amount of time spent on work that could be supported through the adoption of generative AI based on actual time currently spent. Through the development of an analytical tool which focuses on a broad range of tasks undertaken across various areas of the public sector, our approach supports efforts to broaden the adoption of technology beyond the centre of government, and to ensure gains cascade across all service areas.

Our preliminary analysis of the impact generative AI could have on public sector work in Great Britain shows that there is potential for over a third of public sector workers' time to be supported by the adoption of the technology on average. Using the PSTUS dataset, we were able to analyse and group the results around sector and non-sector specific activities to better understand differences across sectors without limiting our analysis to a single area of the public sector. We were also able to explore the differences in generative AI exposure based on whether a public sector worker is a frontliner or not, and whether they are in a managerial or non-managerial position. These findings can help support a more tailored and experiential approach to adoption across service sectors and areas of operation. They illustrate that a blanket, one-size-fits-all approach to adoption across the public sector that ignores these varying dimensions would likely lead to unintended consequences, ranging from missed opportunities in some areas to disorganised uptake in others.

With the above in mind, it is important to briefly discuss what our research does not cover, and where there may be room for future research to expand upon this work. Firstly, our analytical method offers necessary insight into the technical dimensions of generative AI adoption, accounting for the degrees of interpersonal interaction required to complete a given task or activity, as well the degree to which an incorrect or erroneous outcome could result in unintended risks or harms to impacted groups. It does not, however, account for the commercial costs associated with the adoption of new technologies; nor does it explore the various other strategic factors determining successful implementation, such as the availability of internal technical capacity and

expertise, concerns around data security and privacy compliance, or the complexities of integrating new solutions into existing operational structures.

Even so, decisions around generative AI adoption must extend beyond technical – and indeed economic and commercial – feasibility, accounting for the various socio-technical elements that also determine successful adoption. Such decisions must also include considerations of the various operational costs associated with the adoption of state-of-the-art technologies, including the ecological, ethical, and legal impacts (Domínguez Hernández et al. 2024). Further, our method does not account for the more complex and nuanced dimensions of communication, decision-making, and discretion in the public sector (Erdurmazlı 2020; Binns 2022) and does not replace the need to undertake research activities which explore and sufficiently address these interdependencies. Therefore, while these findings serve as valuable datapoints which demonstrate the breadth of potential generative AI application, they should not be viewed independently of the abovementioned considerations. Rather, they should be used to support adoption initiatives alongside existing tools and frameworks such as the [Algorithmic Transparency Recording Standard Hub](#), [data protection impact assessments](#), and the [Generative AI Framework for HMG](#).

Secondly, all the work activities under study were assessed in isolation, not within the broader context of other activities which may build on them. For the purposes of this preliminary study, this view of activities provides helpful groundwork for enabling an understanding of key dimensions which determine exposure to generative AI on a more granular level. However, when considering the application of generative AI within wider workflows, it is important to explore the interdependencies between activities and the impact this may have on feasibility of generative AI adoption. Indeed, when viewed within its specific context, we might find that the same activity could be given a different exposure score. For example, the task ‘reading, reviewing, or editing documentation or paperwork’ might have a lower error tolerance in the context of the ‘Courts and tribunals’ sector than in the context of ‘primary education’, resulting in a lower exposure score for the former. On the other hand, a task like ‘reporting, assessing, or marking pupil’s work’ – which has currently been given a moderately high exposure score (Figure 1) – will still require a nuanced understanding of students, their needs, and their individual circumstances, thereby potentially limiting generative AI’s effectiveness when applied within the teaching context.⁵

It is also worth highlighting that - for the purposes of this pilot survey - the PSTUS did not account for different working patterns over the survey’s reference period. As such, the

⁵ A forthcoming report by the project team at The Alan Turing Institute has found that only 4% of teachers report using generative AI for marking. Here, we can clearly observe the ‘technical feasibility ≠ successful adoption’ principal at play.

results presented in this report are likely to differ when applied to a wider sample that captures more varied working patterns over a longer reference period. Finally, any large-scale adoption of technology like generative AI requires buy-in from its users, and those impacted by its adoption – a growing body of research exists which explores the dimensions that impact trust and preferences in both the public and in public sector workers (Ahn and Chen 2022; Robles and Mallinson 2025; Li, Sun, and Xi 2025). Further in-depth research is required to better capture the factors which shape how the users of the technology - i.e. public sector workers for whom these systems will be integrated into their wider workflows - will navigate questions of trust and understanding.

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Appendix

A. List of work activities provided within the public sector time-use data survey

Below is a list of the activities that were available for public sector workers completing the time-use survey to choose from, broken down by their sector or non-sector specific categorisations. Work activities that were excluded from our analysis are shaded. In total, there are 91 work activities, of which 11 were excluded from our analysis. These categories were taken directly from the official public sector time use statistics published by the Office for National Statistics. The full dataset can be found [here](#).

Code	Activity	Activity grouping	Category
1	Sleeping or resting	Non-work activities	Other
2	Personal care, e.g. washing, getting dressed, eating or drinking	Non-work activities	Other
3	Looking after children	Non-work activities	Other
4	Looking after adults	Non-work activities	Other
5	Household work, e.g. cooking, cleaning, laundry	Non-work activities	Other
6	Volunteering	Non-work activities	Other
7	Entertainment and socialising	Non-work activities	Other
8	Exercise, sports and well-being	Non-work activities	Other
9	Other activities outside of work	Non-work activities	Other
10	Travelling to or from a place of paid work	Other travel	Other
11	Travelling as part of paid work	Travelling as part of paid work	Other
12	Travelling to or from another place	Other travel	Other
13	Teaching or assisting learners in the classroom	Student management	Sector-specific
14	Coaching or assisting learners outside the classroom	Student management	Sector-specific
15	Supervising children outside of the classroom	Student management	Sector-specific
16	Preparing classroom lessons	Student management	Sector-specific
17	Preparing or delivering extra-curricular activities	Student management	Sector-specific
18	Reporting, assessing or marking pupil's work	Student management	Sector-specific
19	Registering attendance	Student management	Sector-specific
20	Safeguarding and wider support for pupils	Student management	Sector-specific
21	Other teaching or education activities	Student management	Sector-specific
22	Assessing patients	Patient management	Sector-specific
23	Observing patients	Patient management	Sector-specific
24	Treating patients	Patient management	Sector-specific
25	Supporting patient care, e.g. changing beds, bathing, feeding	Patient management	Sector-specific

26	Recording medical notes	Records management	Sector-specific
27	Writing medical reports or treatment planning	Records management	Sector-specific
28	Treatment preparation	Other healthcare	Sector-specific
29	Writing prescriptions	Patient management	Sector-specific
30	Other healthcare activities	Other healthcare	Sector-specific
31	Crime-related activities	Policing activity	Sector-specific
32	Non-crime-related activities, e.g. public safety and welfare	Policing activity	Sector-specific
33	Patrolling	Policing activity	Sector-specific
34	Offender-facing activities	Offender management	Sector-specific
35	Non-offender-facing activities	Records management	Sector-specific
36	Other offender management activities	Offender management	Sector-specific
37	Intelligence gathering	Policing activity	Sector-specific
38	Community activities or events	Other policing	Sector-specific
39	Crime prevention activities	Policing activity	Sector-specific
40	Court duties	Court duties	Sector-specific
41	Warrant executions	Policing activity	Sector-specific
42	Prison interviews	Policing activity	Sector-specific
43	Security checks and searches	Offender management	Sector-specific
44	Other policing and public order activities	Other policing	Sector-specific
45	Working directly with members of the public, service users or customers	Other public service duties	Non-sector-specific
46	Social care activities	Other public service duties	Non-sector-specific
47	Environmental or waste management activities	Other public service duties	Non-sector-specific
48	Housing or welfare related activities	Other public service duties	Non-sector-specific
49	Safeguarding	Other public service duties	Non-sector-specific
50	Military duties	Other public service duties	Non-sector-specific
51	Checking, testing, cleaning or replenishing equipment	Other public service duties	Non-sector-specific
52	Other public service activities	Other public service duties	Non-sector-specific
53	Photocopying, printing or scanning	Other general administration	Non-sector-specific
54	Storing, filing or managing documents	Records management	Non-sector-specific
55	Recruitment-related administration	Other general administration	Non-sector-specific
56	Reading, reviewing or editing documentation or paperwork	Records management	Non-sector-specific

57	Booking travel and accommodation	Other general administration	Non-sector-specific
58	Taking notes	Records management	Non-sector-specific
59	Ordering supplies or equipment	Other general administration	Non-sector-specific
60	Finance-related activities	Other general administration	Non-sector-specific
61	Auditing	Other general administration	Non-sector-specific
62	Health and safety	Other general administration	Non-sector-specific
63	Data entry	Records management	Non-sector-specific
64	Creating or updating records, databases, case files or similar documents	Records management	Non-sector-specific
65	Completing or processing forms	Records management	Non-sector-specific
66	Creating or updating an appointment or booking	Records management	Non-sector-specific
67	Organising meetings, meeting administration	Meetings and events	Non-sector-specific
68	Attending a meeting or event	Meetings and events	Non-sector-specific
69	Planning	Project management	Non-sector-specific
70	Writing or reviewing reports	Project management	Non-sector-specific
71	Emails	Communication	Non-sector-specific
72	Phone or video calls	Communication	Non-sector-specific
73	Instant messaging	Communication	Non-sector-specific
74	Having a conversation	Communication	Non-sector-specific
75	Writing letters	Communication	Non-sector-specific
76	Postal or mailing duties	Other general administration	Non-sector-specific
77	Media or press duties	Communication	Non-sector-specific
78	Creating or updating website, noticeboard or display boards	Communication	Non-sector-specific
79	Staff-related administration	Staff management	Non-sector-specific
80	Staff management	Staff management	Non-sector-specific
81	Conducting research	Training and research	Non-sector-specific
82	Personal study	Training and research	Non-sector-specific
83	Attending or completing a course	Training and research	Non-sector-specific
84	Physical or practical training	Training and research	Non-sector-specific
85	Delivering a training course	Training and research	Non-sector-specific
86	Data processing, data analysis, data science	Data processing, data analysis, data science	Non-sector-specific
87	Waiting	Other work activities	Non-sector-specific

88	Taking a work or lunch break	Other work activities	Non-sector-specific
89	Changing clothes	Other work activities	Non-sector-specific
90	Sleeping or resting during work time	Other work activities	Non-sector-specific
91	Other work activity	Other work activities	Non-sector-specific

B. Rubric for assessing the potential of applying generative AI to public sector activities

B1. Introduction

This rubric looks to identify the potential for applying generative AI to both sector and non-sector specific public sector activities. It measures the exposure of these activities to generative AI based on the capabilities of the technology as well as the content of the activity, following the spirit of prior work on quantifying impact of technological change on occupations (Autor, Levy, and Murnane 2003) and potential suitability of supervised machine learning to occupations (Brynjolfsson, Mitchell, and Rock 2018).

The rubric is a combination of two ‘automation exposure’¹ rubrics developed by Eloundou et al. to measure the impact that direct access to generative AI could have on the quality and completion time of various tasks (Eloundou et al. 2024). We follow Eloundou et al.’s design of the rubric, adjusting the text in order to reflect the application of the rubric to a public sector context. We fully make use of their list of generative AI capabilities, with no changes made to the text. While Eloundou et al., use the term ‘automation’ to refer to the impact of generative AI use, we remove references to automation and instead use language which refers to the supporting capabilities of generative AI.

Additionally, we also follow the comprehensive taxonomies of tasks for assessing the impact of new technologies of work (Fernández-Macías and Bisello 2022) by also considering the methods and tools of work, which refer to (1) the physical, material methods and tools that may be undertaken as part of the completion of an activity, and (2) the way an activity may be organised around social interaction.

B2. Rubric assessment statement

Consider the most powerful current generative AI models (such as the language model GPT-4). These models can complete many tasks that can be formulated as having text, audio or image input and output where the context for the input is typically captured in a short prompt text statement.

Assume you are a public sector worker with an average level of expertise in your role trying to complete a given activity. You have access to the AI model via an API or web interface. You also have access to any commonly available technical tools accessible via a laptop (e.g., a microphone, speakers, etc.). You do not have access to any other physical tools or materials. Please label each of the activities according to the capabilities and rubric below.

B3. Capabilities

On its own, the generative AI model can perform tasks that can be reduced to:

- Writing and transforming text and code according to complex instructions,
- Providing edits to existing text or code following specifications,
- Writing code that can help perform simple, repetitive digital tasks,
- Debugging code or software
- Programming in computer languages like Python, C++, Java, Stata
- Assisting with a data analysis
- Translating text between languages,
- Summarizing short documents,
- Providing feedback on documents,
- Answering questions about a document,
- Generating questions a user might want to ask about a document,
- Writing questions for an interview or assessment,
- Writing and responding to emails, including ones that involve refuting information or engaging in a negotiation (but only if the negotiation is via written correspondence),
- Maintain records of written data,
- Prepare training materials based on general knowledge,
- Searching for a relevant document over a small number of documents,
- Reading text from PDFs,
- Scanning images,
- Creating or editing digital images according to instructions,
- Captioning an image,
- Transcribing text, or

- Informing anyone of any information via any written or spoken medium.

With additional software built on-top of the generative AI model, its capabilities grow to include:

- Summarizing documents longer than 2000 words and answering questions about those documents,
- Retrieving up-to-date facts from the Internet and using those facts in combination with the generative AI capabilities,
- Searching over an organisation's existing knowledge, data, or documents and retrieving information,
- Retrieving highly specialized domain knowledge,
- Making recommendations given data or written input,
- Analysing written information to inform decisions,
- Preparing training materials based on highly specialized knowledge,
- Performing legal research,
- Providing counsel on issues,
- Writing lesson plans, and
- Maintaining databases

Definitions:

- **Equivalent quality** means someone reviewing the output of this activity would not be able to tell whether a human completed it on their own or with assistance from an AI system,
- **Digital information** - or information that can easily be expressed digitally- includes but is not limited to text, audio, images, PDFs, books, and code.

B4. Rubric scoring guide

Score	Description
<i>No Exposure (E0)</i>	Label activities as no exposure (E0) if a state-of-the-art generative AI model cannot conceivably perform any aspect of this activity in any manner. Activities labelled as E0 are likely to: - Require physical manipulation and transformation of material objects and/or, - Involve a high degree of interpersonal interaction - Have a low error tolerance – i.e., consequences for failures or inaccuracies are serious
<i>Full Exposure (E4)</i>	Label activities as full exposure (E4) if a state-of-the-art generative AI model can fully execute this activity with equivalent expected quality of output and a significant reduction in time spent completing the activity. Activities labelled as E4 are likely to: Not require manipulation and transformation of material objects and/or, - Have low interpersonal interaction - Rely exclusively on the processing of digital information - Have a high error tolerance – i.e., consequences for failures or inaccuracies are small - Not require human oversight
<i>High Exposure (E3)</i>	Label activities as high exposure (E3) if a state-of-the-art generative AI can execute the majority of this activity with equivalent expected quality of output and a significant reduction in time spent completing the activity. If the generative AI model is not able to achieve this alone, it is easy to imagine additional software that could be developed on top of the model that would allow it to fully execute this activity. Activities labelled as E3 are likely to: Not require manipulation and transformation of material objects, - Have low or medium interpersonal interaction, - Rely almost exclusively on the processing digital information - Have a reasonable error tolerance – i.e. consequences for failures or inaccuracies are significant - Some human oversight is required

<i>Moderate Exposure (E2)</i>	Label activities as moderate exposure (E2) if a state-of-the art generative AI can execute portions of this activity with equivalent expected quality of output and a reduction in time spent completing the activity. If the generative AI model is not able to achieve this alone, it may be possible to imagine additional software that could be developed on top of the model that would allow it to execute more of this activity. However, activities labelled as E2 involve actions that a state-of-the-art generative AI model cannot execute to equivalent expected quality - whether alone, or with additional software. Activities labelled as E2 are likely to: - Require some manipulation and transformation of material objects, - Have medium or high interpersonal interaction, - Rely heavily on the processing digital information - Have low error tolerance – i.e. consequences for failures or inaccuracies are significant - Require human oversight
<i>Low Exposure (E1)</i>	Label activities as low exposure (E1) if a state-of-the art generative AI cannot execute the majority of the activity and may only be able to support with smaller actions. Activities labelled as E1 therefore involve actions that a state-of-the-art generative AI model cannot execute to equivalent expected quality - whether alone, or with additional software. Activities labelled as E1 are likely to: - Require manipulation and transformation of material objects, - Have medium or high interpersonal interaction, - Rely partially on the processing digital information - Have low error tolerance – i.e. consequences for failures or inaccuracies are significant - Require human oversight – the majority of actions in this activity will be executed by humans

B5. Numerical values

Following the authors of the rubric, the scores should be numerically expressed as follows:

- E0 is assigned a score of 0
- E1 is assigned a score of 0.25
- E2 is assigned a score of 0.5
- E3 is assigned a score of 0.75
- E4 is assigned a score of 1